INCH-POUND
MIL-DTL-83503/7D
17 June 1998
SUPERSEDING
MIL-C-83503/7C
1 July 1983

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, FLAT CABLE, NONENVIRONMENTAL PLUG WITH DUAL POLARIZATION, ROUNE\D, INSULATION DISPLACEMENT NON-REMOVABLE SOCKET CONTACTS (.100 X .100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-83503.

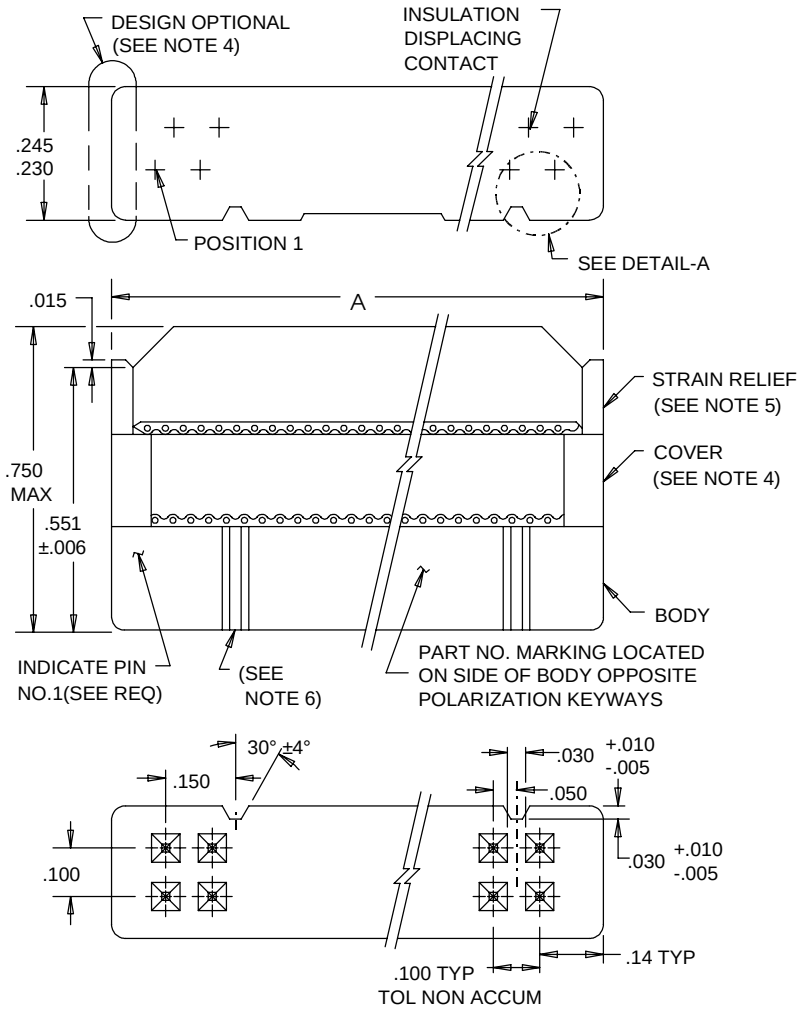

Figure 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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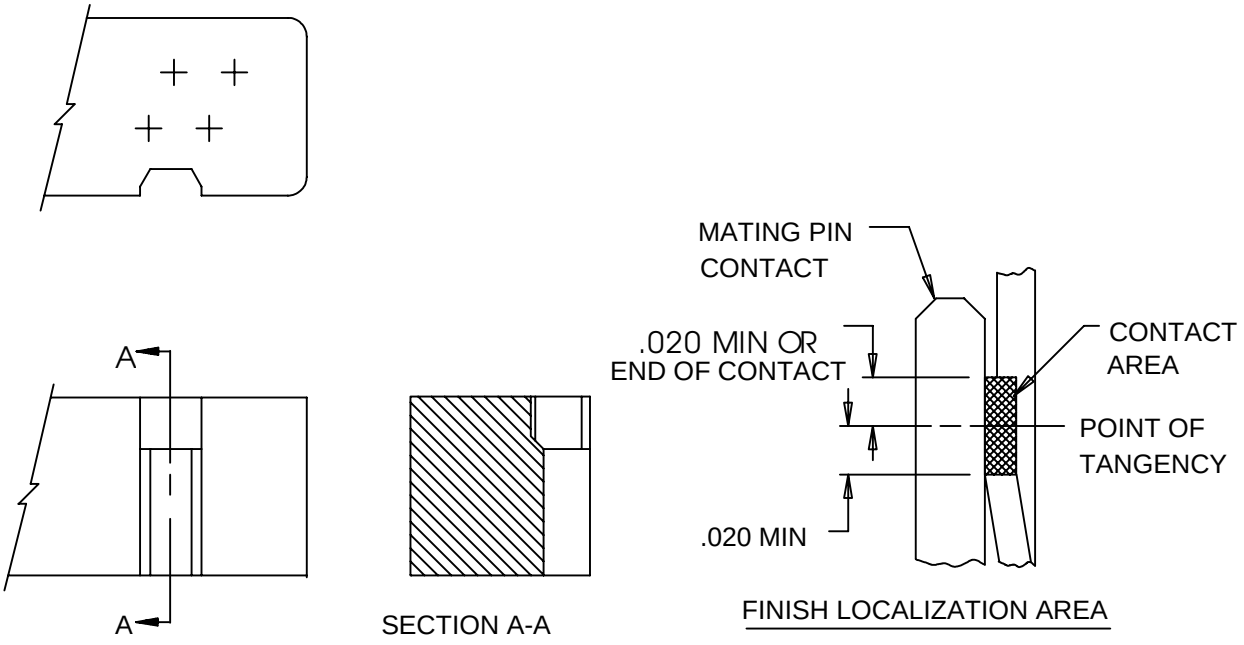

DETAIL

ALTERNATE POLARIZATION NOTCH
WITH NON FUNCTIONAL DETENT

Inches	mm	Inches	mm
.005	0.13	.100	2.54
.006	0.15	.14	3.6
.010	0.25	.150	3.81
.015	0.38	.230	5.84
.020	0.51	.245	6.25
.030	0.76	.551	14.00
.050	1.27	.725	18.42

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.01 inch (0.3 mm) for two place decimals and ±.005 inch (0.13 mm) for three place decimals and 2° on angles.
4. Strain relief shall be provided. Strain relief design is optional (may be either separate of an integral part of cover).
5. Assembled height including installed cable and (when applicable) the strain relief and excluding leads shall be .40 inch (10.2 mm) maximum.
6. Tolerances are non-cumulative.

FIGURE 1. Dimensions and configuration - Continued.

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TABLE I. Dash numbers and characteristics

Dash number	Number of contacts	Dimensions 'A inches 1/	Mating Connector	
			Header (straight) M83503/21- or M83503/25-	Header (right angle) M83503/20- or M83503/24-
01	10	.68 (17.3)	01, 02. 03	01, 02, 03
02	14	.88 (22.4)	04, 05, 06	04, 05, 06
03	16	.98 (24.9)	07, 08, 09	07, 08, 09
04	20	1.18 (30.0)	10, 11, 12	10, 11, 12
05	24	1.38 (35.1)	-------	-------
06	26	1.48 (37.6)	13, 14, 15	13, 14, 15
07	30	1.68 (42.7)	-------	-------
08	34	1.88 (47.8)	16, 17, 18	16, 17, 18
09	40	2.18 (55.4)	19, 20, 21	19, 20, 21
10	50	2.68 (68.1)	22, 23, 24	22, 23, 24
11	60	3.18 (80.8)	25, 26, 27	25, 26, 27
12	64	3.38 (85.9)	28, 29, 30	28, 29, 30

REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figure 1 and table I.

Temperature range: -55° C to +120° C.

Voltage rating: 300 V rms.

Current rating: 1 ampere.

Materials:

Contacts: Beryllium copper in accordance with ASTM B196 or ASTM B197, phosphor bronze in accordance with ASTM B139. Gold plating .000050 minimum shall be in the contact mating area (see MIL-DTL-83503).

Housing, cover, and strain relief: Glass reinforced polyester in accordance with MIL-M-24519, Type GPT-15F, GPT-20F, or GPT-30For an approved equivalent material manufactured to other commonly accepted industry standards. Cover retaining device may be made of QQ-S-763 class 301 stainless steel or an approved equivalent material manufactured to a commonly accepted industry standard.

Conductor accommodation: 28 AWG stranded .05 inch spacing between conductor centerlines.

Test wire size: 28 AWG stranded.

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Dielectric withstanding voltage:

(at sea level): 500 V rms.

(at altitude): 200 V rms.

Contact resistance: 50 milliohms, maximum.

Interface:

Cable: Use round conductor flat cable in accordance with MIL-C-49055/1 and /11 through /13, DESC drawing 92011. Overall cable thickness shall not exceed .042 inches

Connector interface: These connectors are designed to mate with connectors defined on MIL-DTL-83503/20, /21, /24, /25, or .025 inch (round or square) posts with a .230 inch minimum length. (NOTE: These connectors are not intended to be stacked side by side on a .100 X .100 inch pin field.)

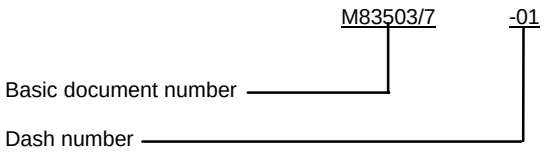
Marking:

Contact identification: Molded triangle “A” or notch to show number 1 contact or raised numbers to show position number “1”.

Marking:

Part or identifying number (PIN): The PIN shall consist of this specification sheet and the dash number from table I.

Example



NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians:
Air Force - 85

Preparing activity:
DLA - CC

(Project 5935-4110-02)

Review Activities:
Air Force - 17, 99

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